

B.E Sem VIII Comp e - Scheme - R - 2019 Aug 25

(3 Hours)

Total Marks 80

21/8/25

NB:

- 1) Question **number 1** is compulsory
- 2) Attempt **any three** out of the remaining **five questions**.
- 3) Assume suitable data if **necessary** and justify the assumptions.
- 4) Figures to the **right** indicate full marks

Q1 Attempt any four 20

- A Explain in brief the data science process
- B Compare and contrast descriptive and inferential statistics
- C Explain in brief, Univariate Visualization and Multivariate Visualization
- D Explain k-fold cross-validation and leave-one-out cross-validation.
- E Discuss briefly the taxonomy of time series forecasting techniques.

Q2 A Explain Data Science tasks 10

B Explain the working of the ARIMA model in detail 10

Q3 A What is Data Visualization? Explain with an example each of the following: 10
i) Histogram, ii) Quartile, and iii) scatter plot

B State different Simple Forecasting Methods? Explain Moving Average Smoothing and Weighted Moving Average Smoothing 10

Q4 A Explain the Distance-based approach to outlier detection. 10

B Calculate the Bowley's coefficient of skewness from the following data 10

Income in Rs	No of Person	Income in Rs	No of Person
300-400	14	800-900	62
400-500	46	900-1000	48
500-600	58	1000-1100	22
600-700	76	1100-1200	06
700-800	68		

Q5 A Explain the different components of Time series decomposition. 10

B Explain the roadmap for data Exploration. 10

Q6 A Explain the DBSCAN algorithm to detect outliers. Give the advantages and disadvantages of the algorithm 10

B Calculate Karl Pearson's correlation coefficient between the marks in English and Hindi obtained by 10 students 10

Marks in English	10	25	13	25	22	11	12	25	21	20
Marks in Hindi	12	22	16	15	18	18	17	23	24	17